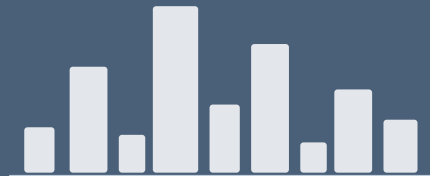


Interaction of a cumulus cloud ensemble with the large-scale environment, Part I

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How do a spectrum of cumulus clouds and the large-scale flow control each other, and how can the clouds' collective effect be predicted?



The problem: parameterize a cloud ensemble

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Cumulus parameterization

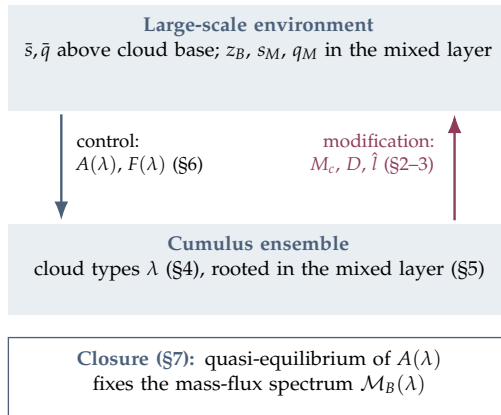
Many cumulus clouds live inside one large-scale disturbance, on much smaller scales. Predict only their collective effect, from large-scale variables.

Goal

A closed theory of the two-way interaction, with the subcloud mixed layer included.

Before this paper

CISK (Charney & Eliassen 1964; Ooyama 1964): too crude for general use. Ooyama (1971): a spectrum of non-interacting bubbles, with the “dispatcher function” left open.



Closure: the clouds keep $A(\lambda)$ in balance with the large-scale forcing

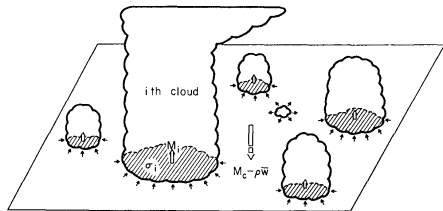


Fig. 1: a unit area between cloud base and the highest top. Tall clouds entrain; a cloud that lost buoyancy detrains.

$$M_i = \rho \sigma_i w_i, \quad M_c = \sum_i M_i, \quad \sigma_c = \sum_i \sigma_i \ll 1$$

$\tilde{M} = \rho \bar{w} - M_c$: subsidence between the clouds

$$s = c_p T + g z, \quad h = s + L q$$

Environment above the mixed layer

$$\rho \partial_t \bar{s} = D(\hat{s} - \bar{s} - L \hat{l}) + M_c \partial_z \bar{s} - \rho \bar{v} \cdot \nabla \bar{s} - \rho \bar{w} \partial_z \bar{s} + \bar{Q}_R$$

$$\rho \partial_t \bar{q} = D(\hat{q}^* + \hat{l} - \bar{q}) + M_c \partial_z \bar{q} - \rho \bar{v} \cdot \nabla \bar{q} - \rho \bar{w} \partial_z \bar{q}$$

Detrainment: saturated cloud air with liquid water spreads out; its evaporation cools and moistens.

Cumulus-induced subsidence: $M_c \partial_z \bar{s} > 0$ warms, $M_c \partial_z \bar{q} < 0$ dries.

Large scale: advection, mean vertical motion, radiation.

Condensation heating stays in the clouds. It keeps them buoyant and sustains M_c , so the environment warms through the subsidence, not directly.

The ensemble acts on the environment only through $M_c(z)$, $D(z)$ and $\hat{l}(z)$

Cloud type λ : fractional entrainment rate, constant with height. Since $\mu = 2\alpha/R$ (Simpson), large λ means small clouds.

$$\partial_z \eta = \lambda \eta \Rightarrow \eta(z, \lambda) = e^{\lambda(z-z_B)} \text{ up to } z_D(\lambda)$$

$$h_c(z, \lambda) = \frac{1}{\eta(z, \lambda)} \left[h_M + \lambda \int_{z_B}^z \eta(z', \lambda) \bar{h}(z') dz' \right]$$

Top $z_D(\lambda)$: buoyancy vanishes, $h_c = \hat{h}^* \approx \bar{h}^*$.

$$M_c(z) = \int_0^{\lambda_D(z)} \mathcal{M}_B(\lambda) \eta(z, \lambda) d\lambda$$

$$D(z) = -\mathcal{M}_B(\lambda_D) \eta(z, \lambda_D) \frac{d\lambda_D}{dz}$$

$\lambda_D(z)$: the type whose top is at z .

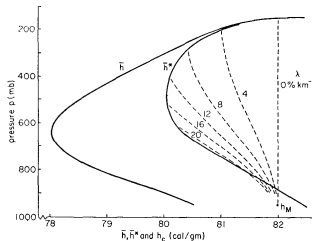


Fig. 6: $\bar{h}$, $\bar{h}^*$ and h_c for $\lambda = 0-20\% \text{ km}^{-1}$; tropical sounding.

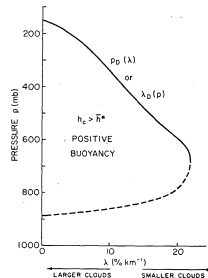
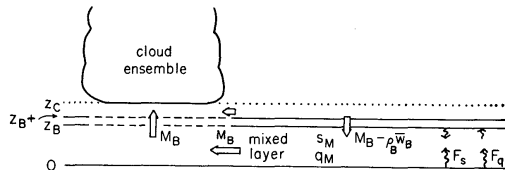


Fig. 8: top $p_D(\lambda)$.

In the paper

A tower of constant radius in the Lagrangian sense fits observations better than expanding thermals or starting plumes (Simpson et al. 1965).

Given the environment, each cloud type is fixed; only $\mathcal{M}_B(\lambda)$ is unknown



Well mixed for $0 < z < z_B$ (s_M , q_M , h_M), with jumps Δs , Δq at z_B , about 200 m below cloud base z_C . Clouds root here: $h_c(z_B, \lambda) = h_M$.

$$\rho_B \frac{Dz_B}{Dt} = -(M_B - \rho_B \bar{w}_B) + k \frac{(F_{sv})_0}{\Delta s_v}$$

Subsidence lowers z_B ; the surface buoyancy flux deepens it ($k \approx 0.1$ – 0.25). A shallower layer feeds less mixed-layer air to the clouds, so z_B helps control M_B .

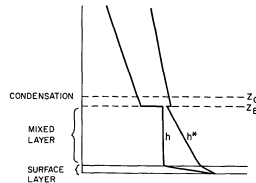


Fig. 9: mixed layer below the clouds; jumps at z_B .

Fig. 10: h is uniform in the mixed layer; h^* falls with height (dry adiabatic).

In the paper

Below cloud base, the mass flux is carried by updrafts tied to the clouds, unsaturated between z_B and z_C .

A trigger is needed; it may act on z_B rather than on s_M , q_M . If z_B is well below condensation, small turbulent perturbations rarely trigger clouds; new ones form in the wake of earlier ones.

The mixed layer feeds the clouds, and its depth is part of the control

$$\frac{d\mathcal{K}(\lambda)}{dt} = A(\lambda) \mathcal{M}_B(\lambda) - \mathcal{D}(\lambda)$$

$$A(\lambda) = \int_{z_B}^{z_D(\lambda)} \frac{g}{c_p \bar{T}(z)} \eta(z, \lambda) [s_{vc}(z, \lambda) - \bar{s}_v(z)] dz$$

Kinetic energy generated per unit cloud-base mass flux: an η -weighted buoyancy integral.

Fixed by the profiles of the environment, mixed layer included: the only environmental property that controls the kinetic energy of type λ .

$\lambda = 0$: undiluted ascent, insensitive to humidity above cloud base. Larger λ also needs a moist environment ($\bar{h}^* - \bar{h}$ small).

$A > 0$ is necessary, not sufficient: $\mathcal{M}_B > 0$ and a trigger are also needed.

$A(\lambda)$ extends moist convective instability to each cloud type

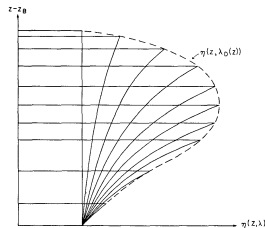


Fig. 5: $\eta(z, \lambda)$ for several λ ; the envelope is $\eta[z, \lambda_D(z)]$.

Proposal, not in the paper

For $\lambda = 0$, A is an undiluted-parcel buoyancy integral, close to what is now called CAPE.

What changes $A(\lambda)$: clouds and large-scale forcing

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$$\frac{dA(\lambda)}{dt} = \underbrace{\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda'}_{\text{clouds}} + \underbrace{F(\lambda)}_{\text{large scale}}$$

Large-scale forcing $F = F_C + F_M$

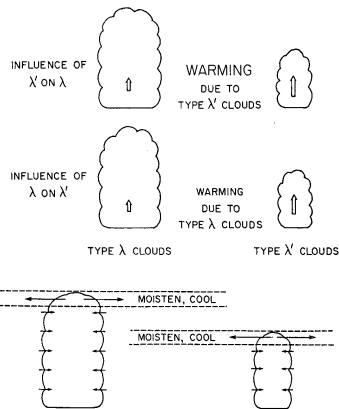
Positive F destabilizes. F_C : mainly cooling by large-scale ascent above the mixed layer. F_M : mixed-layer deepening by ascent and surface fluxes.

Kernel $K = K_V + K_D + K_M$

K_V : subsidence warming by type λ' lowers $A(\lambda)$; usually negative, nearly symmetric (Fig. 11).

K_D : zero for $\lambda' < \lambda$, mostly positive for $\lambda' > \lambda$: shallower clouds moisten and cool the environment of deeper ones (Fig. 12).

K_M : cloud-induced changes of the mixed-layer depth.



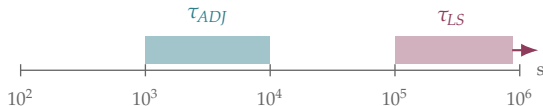
The clouds typically stabilize ($K < 0$) what the large scale destabilizes ($F > 0$)

Two time scales

τ_{ADJ} : clouds remove A . τ_{LS} : the large scale changes.

$$\left| \left[\frac{dA}{dt} \right]_C \right| \sim \frac{A}{\tau_{ADJ}}, \quad \left| \frac{dA}{dt} \right| \lesssim \frac{A}{\tau_{LS}}$$

$A \sim (10^{-1} \text{ m s}^{-2}) H_c$ and $[dA/dt]_C \sim (10^{-4} - 10^{-5} \text{ m s}^{-3}) H_c$ give $\tau_{ADJ} \sim 10^3 - 10^4$ s, against $\tau_{LS} \gtrsim 10^5$ s.



Closure

If $\tau_{ADJ} \ll \tau_{LS}$, dA/dt is negligible against the cloud term:

$$\int_0^{\lambda_{\max}} K(\lambda, \lambda') \mathcal{M}_B(\lambda') d\lambda' + F(\lambda) = 0$$

A Fredholm equation of the first kind for $\mathcal{M}_B \geq 0$; where $\mathcal{M}_B(\lambda) = 0$, the left side is negative.

A sequence of quasi-neutral states: A stays small, not constant; compare quasi-geostrophic balance.

Only $\mathcal{M}_B = \mathcal{N} m_B / \tau$ is needed, not population, lifetime and single-cloud mass flux separately.

The cloud spectrum is the one that balances the large-scale forcing

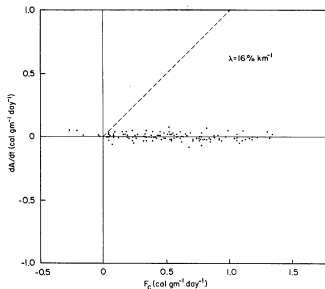
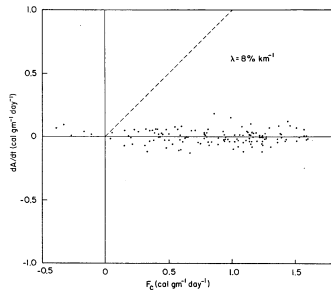


Fig. 13: observed dA/dt against F_C for $\lambda = 8$ and $16\% \text{ km}^{-1}$; dashed: $dA/dt = F_C$, the change without clouds.

Marshall Islands data (Yanai et al. 1973): $dA/dt \approx 0$ while F_C reaches $1.5 \text{ cal g}^{-1} \text{ day}^{-1}$. I_1 and I_2 swing by about 20 cal g^{-1} , but $A = I_1 - I_2$ stays small and positive. F_M was not computed; a rough estimate makes it much smaller than F_C .

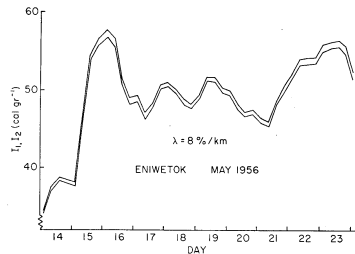


Fig. 14: I_1 and I_2 at Eniwetok, May 1956, for $\lambda = 8\% \text{ km}^{-1}$; $A = I_1 - I_2$.

In tropical data, $A(\lambda)$ stays near neutral while the forcing varies

Limits

Built into the theory

One parameter λ per cloud type; entraining updrafts only, no downdrafts.

Detrainment only at cloud top; crude rainfall, $r = C_0 l$.

$\sigma_c \ll 1$; horizontally uniform environment; unsaturated mixed layer.

Only $\mathcal{M}_B(\lambda)$: no population or lifetime.

Trigger assumed, not modelled; interactions with shear neglected.

Evidence

One tropical oceanic data set, without F_M ; the solution method is deferred to Part II.

Our reading [ours]

The closure needs $\tau_{ADJ} \ll \tau_{LS}$: fast forcing, such as daytime heating over land or mountains, is not tested here.

Proposal, not in the paper

1. Soaring-day index. $A(\lambda)$ and z_B from soundings or reanalysis for each flight day; relate them to climb rates, climb tops and H .

2. Climb tops as cloud tops. A dry entraining plume gives $z_D(\lambda)$ from the sounding; where climbs end would then map to thermal types.

3. Thermal ceiling. Eq. (126) predicts the rise of z_B from surface fluxes; compare it with the daily rise of climb tops.

Link. Neggers et al. (2003) measure the mass flux by cloud size in LES: an empirical view of the unknown $\mathcal{M}_B(\lambda)$.

A closed statistical theory of convection; its links to thermals are ours to test